

Optimal short-term coordination of water-heat-power nexus incorporating plug-in electric vehicles and real-time demand response programs



Farkhondeh Jabari^a, Hamid Jabari^b, Behnam Mohammadi-ivatloo^{a,*}, Jafar Ghafouri^b

^a Faculty of Electrical and Computer Engineering, University of Tabriz, Tabriz, Iran

^b Department of Mechanical Engineering, Tabriz Branch, Islamic Azad University, Tabriz, Iran

ARTICLE INFO

Article history:

Received 16 March 2018

Received in revised form

12 February 2019

Accepted 17 February 2019

Available online 20 February 2019

Keywords:

Water-heat-power (WHP) nexus

Combined water and power (CWP) plants

Combined heat and power (CHP) units

Plug-in electric vehicles (PEVs)

Demand response programs (DRPs)

ABSTRACT

In recent years, the population growth has caused the water and energy shortages. Therefore, the current paper proposes a novel method for day-ahead optimization of integrated water-heat-electricity systems with the aim of minimizing the fuel cost of the desalination, heat treatment and power production units. The technical limits of the power only units, combined power and water generation units, cogenerators of heat and electricity, heat only units, and the seawater purification process are incorporated in optimization problem. In addition, a real-time demand side management strategy is applied on heat, pure water and power loads to reduce the objective cost function as low as possible by shifting a percentage of each demand from peak periods to mid or off peak hours. Moreover, aggregators of pure electric vehicles participate in bulk energy management by charging electricity during off-peak low-price hours and discharging it over the on-peak high-price periods. To validate the feasibility and cost-effective performance of the water-heat-power hub system, a mixed-integer non-linear program is developed by generalized algebraic mathematical modelling system (GAMS) optimizer and solved by branch-and-reduce optimization navigator (BARON) tool. The ramp down and ramp up limits of the thermal power plants, generation capacity of water/heat/power producers, and the load-generation balance criterion are considered as optimization constraints. It is found that \$659,573 cost saving is obtained due to participation of PEVs and DRPs in trigeneration process.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

Under the hot weather condition, the consumption of the fresh water and electricity increases. In south of Iran, a huge value of seawater is available. But, the water scarcity and the electricity cascaded outages occurs in summer. Because, the seawater desalination plants and the power generation units should optimally be co-dispatched and the interdependence of the power procurement and water purification processes should be considered [1]. Meanwhile, population growth has caused the transmission scale power systems are operated in stability boundaries [2,3]. If the uncertainties of the water/heat/electricity demands are not considered in short-term or day-ahead scheduling of trigeneration hub

networks, the load-generation imbalance may occur and led to water shortages and wide-spread power outages. Hence, some preventive actions such as forced interrupts [4,5] and incentive based demand response programs (DRPs) [6,7] can be implemented on residential [8], industrial [9,10], and commercial [11] end users. Another way to avoid from the water and energy crises is to use the energy storages such as aggregated electric vehicles [12].

Recently, researchers proposed the interesting methods for designing and co-optimizing the interconnected water and power hub systems. In Ref. [13], the authors investigated the economic performance of the wind powered water-energy hybrid network by determining the amount of the surface/embodyed/groundwater requirement for generating 1 kW electricity and the value of the energy consumption in gross water desalination process. The use of the surplus electricity production of wind farm for salted water purification cycle results in a significant cost saving. Moreover, the fluctuations of the wind speed and power could be modelled in long-term (1 year) operating problem. In Ref. [14], the coupled

* Corresponding author.

E-mail addresses: f.jabari@tabrizu.ac.ir (F. Jabari), hamid.jabari96@yahoo.com (H. Jabari), bmohammadi@tabrizu.ac.ir (B. Mohammadi-ivatloo), jafarghafouri@iaut.ac.ir (J. Ghafouri).

Nomenclature

Indices

t	Time [hour]
g	Thermal generators
w	Desalination unit
c	CWP producer
j	CHP plant
k	Heat only unit

Water-heat-power nexus

C_p	Fuel cost of thermal units [\$]
C_{cwp}	Fuel cost of CWP plants [\$]
C_w	Fuel cost of seawater purifier [\$]
C_{chp}	Fuel cost of CHP plants [\$]
C_h	Fuel cost of boilers [\$]
$P_{g,t}$	Power generation of thermal unit g [MW]
$P_{c,t}$	Power generation of CWP plant c [MW]
$P_{j,t}^{chp}$	Electricity output of CHP producer j [MW]
$W_{c,t}$	Fresh water generation of CWP plant c [m^3/h]
$W_{w,t}$	Pure water production of desalination plant w [m^3/h]
$Q_{k,t}$	Heat output of boiler k [MW]
$Q_{j,t}^{chp}$	Heat generation of CHP plant j [MW]
a_g, b_g, c_g	Fuel cost factors for thermal generation station g
$\alpha_c, \beta_c, \gamma_c, \zeta_c, \xi_c$	Fuel cost factors for CWP plant c
a_k, b_k, c_k	Fuel cost factors for boiler k
$a_j^{chp}, b_j^{chp}, c_j^{chp}, d_j^{chp}, e_j^{chp}, f_j^{chp}$	Fuel cost factors for CHP producer j
$U_{g,t}$	Binary variable, that is 1 if thermal power plant g is on, else it will be 0
$U_{c,t}$	Binary variable, that is 1 if CWP unit c is on, else it will be 0
U_j^{chp}	Binary variable, that is 1 if CHP producer j is on, otherwise it will be 0
$U_{k,t}$	Binary variable, that is 1 if boiler k is on, else it will be 0
a_w, b_w, c_w	Fuel cost factors of desalination plant w
P_g^{min}, P_g^{max}	Min and max of power produced by thermal power plant g [MW]

P_c^{min}, P_c^{max}	Min and max of power produced by CWP plant c [MW]
W_c^{min}, W_c^{max}	Min and max of pure water produced by CWP plant c [m^3/h]
W_w^{min}, W_w^{max}	Min and max of fresh water generated by desalination plant w [m^3/h]
Q_k^{min}, Q_k^{max}	Min and max of heat output of boiler k [MW]
R_c^{min}, R_c^{max}	Min and max of power-water ratio for CWP unit c [MWh/ m^3]
E_t	Energy demand [MW]
W_t	Pure water demand [m^3/h]
Q_t	Heat load [MW]

Demand response programs

$Load^t$	Potable water/heat/electrical demands after applying DRPs
$Load^{t_0}$	Initial values of fresh water/heat/electrical loads before applying DRPs
DR^t	Percentage of water/heat/electrical load reduction
inc^t	Value of water/heat/electrical load increase
DR_{max}	Max percentage of water/heat/electrical load decrease
inc_{max}	Max value of water/heat/electrical load increase

Electric vehicles

$E_{t,i}^{PEV}$	Energy consumption of pure electric vehicle i [kW]
$P_{t,i}^{PEVch}, P_{t,i}^{PEVdis}$	Electrical power charged or discharged by electric vehicle i [kW]
$P_{PEVch}^{min}, P_{PEVch}^{max}$	Min and max power charge [kW]
$P_{PEVdis}^{min}, P_{PEVdis}^{max}$	Min and max power discharge [kW]
$SOC_t^{PEV_i}$	State of charge of vehicle i
$SOC_{PEV_i}^{min}, SOC_{PEV_i}^{max}$	Min and max of SOC of battery pack i
$U_{t,i}^{PEVch}, U_{t,i}^{PEVdis}$	Binary variable, “1” if vehicle i is charged/discharged, else 0
χ, η	Efficiency of charge and discharge modes

water and energy network of Jordan is assessed from technical and political viewpoints. Some key parameters such as physical distance and interconnection of water and power distribution systems and dominant stakeholders of power and water policy makers are merged. The increase of the water/energy efficiency in domestic applications has not the considerable economic benefit. More energy saving can be achieved by reducing the water leakage, replacing the old pumps with new ones, and using the renewable energy sources such as solar and wind for driving the water pumps. Scholars of [15] presented a global scale water-energy trade model based on ecological network analysis (ENA). The ENA algorithm models the trajectories of water and electricity trades for dominant competitors such as China, South and Central America, Middle East, Africa, and Australia. Studies show that China is equal to USA in utility mutual relationships. The synergism factors and the grid mutualism proves that the economic decisions mitigate the potential effects of the global electricity trade in solving the water shortage of China. Reference [16] developed a bi-level water-power

management (BLWPM) program for minimizing the operation cost of the wastewater treatment and power generation plants. The optimization problem is restricted by the mass balance constraint, limitation of fossil fuels, load-generation balance, production capacity, electricity consumption in desalination, fresh water consumption in power plants, limited access to water sources, and emissions of pollutant gases. The BLWPM approach does not consider the site and the water shedding in the distribution scale. Although, the treatment of the waste water has a significant economic impact on combined water and power generation systems, but there is no gross water management strategy in BLWPM model.

In Ref. [17], a water-food-power hub system is studied over a 1-year period. In this model, the grey water is treated and reused in household sector. Moreover, the anaerobic digestion of the waste food for generating the biofuels is used for power generation. The water and electricity demands depends on seasonal conditions, while the waste food production and the food demand is less sensitive to weather variations. The probability of the water

shortage is reduced by reusing the treated waste water for non-potable applications such as agricultural sector. In addition, the geographical location, food habits and culture should be discussed. In Ref. [18], the risk of the power curtails under the low river water flow and the high ambient temperature is minimized by smart utilizing the desalination systems and hydrothermal units. When the mass flow rate of the river water is low, the small increase in temperature may lead the conventional hydrothermal power plants shutdown. Hence, the load-generation mismatch may cause the voltage drop in the power system and cascading failures and catastrophic blackouts. The vulnerability of the hydrothermal units could be assessed in tropical regions. Authors of [19] introduced a policy-constrained model for cogeneration of water and power from river basin hosting hydrothermal units. The abnormal mass flow rate and temperature of the river water has more effect on electricity production of hydrothermal units. The river water could be used for cooling the pumps, transformers, motors, and turbines and increasing the system efficiency. Researchers of [20] optimized the short-term generation schedule of the combined heat and power systems in low-income and stand-alone networks. At off-peak time intervals, the additional heat and power can be used for generation and distribution of pure water.

According to reviewed works, there are many interesting studies on co-optimization of water and power generation systems. But, there is no research on day-ahead economic dispatch of water-heat-power hybrid systems, which consist of thermal power plants, heat only units, co-producers of heat and power, desalination processes, and co-generators of power and potable water. Moreover, participation of aggregated plug-in electric vehicles in water-heat-power nexus has not been studied, yet. As we know, pure electric vehicles with no need to fossil fuel consumption can be used for saving electricity at non-driving and off-peak demand periods and discharging it at plugged and on-peak electricity consumption hours to reduce the output power and fuel cost of the fossil fuels based thermal units and co-generators of heat and power (CHP). The changes of the operating points of CHP and power only units affect the optimum operation of the combined water and power (CWP) generation units. Hence, optimum operating points of desalination plants will vary because they are simultaneously co-dispatched with CWP producers to satisfy the fresh water demand. Meanwhile, the optimum operating points of the boilers, which should be co-scheduled with CHP plants to satisfy the heat demand, will be changed. In other words, incorporation of aggregated electric drive vehicles in energy procurement process will change the optimum operating points of all generation units in water-heat-power tri-generation systems. In addition, demand response programs, which shift a part of heat/water/electrical demands from peak fuel cost periods to less cost hours, have not been applied on integrated water-heat-electricity production systems. Therefore, this paper presents a mathematical model for optimal co-scheduling of water, heat and power producers. The novelties of the current paper consist of:

- Day-ahead optimal co-scheduling of water, heat and power systems is carried out. The fuel cost of the seawater desalination plant, heat extraction and power generation units is simultaneously minimized. The technical limits of the desalination plants, co-generation units, thermal stations, and heat only unit are considered as optimization constraints.
- Optimal charge and discharge decisions of plug-in electric vehicles (PEVs) are scheduled over a 24-h time interval for cost-effective energy management in both water-heat-power nexus and transportation sector. Participation of PEVs in power generation facilities by charging electricity through batteries during off-peak and mid-peak hours and extracting it for satisfying on-

peak demand, reduces emissions of greenhouse gases and ozone-depletion substances, significantly.

- A real-time demand response programming strategy is implemented on water, heat and electricity loads for peak clipping, valley filling and load curve smoothing by shifting a part of each load from on-peak hours to mid-peak and off-peak periods.
- Four cases are studied to participate PEVs and implement DRPs on day-ahead economic coordination of water desalination units, power generation systems and heat treatment cycles.

The remainder of the current paper is structured as follows: Section 2 presents the problem formulation. Afterwards, numerical result and analysis are provided in Section 3. Finally, Section 4 concludes the paper.

2. Problem formulation

2.1. Water-heat-power grid

The optimal short-term scheduling of the combined potable water, heat and electricity production systems consisting of desalination only processes, combined water and power (CWP) plants, heating cycles, combined heat and power (CHP) producers, and thermal generating units are presented as shown in Fig. 1. The electrical, heat and potable water demands of the test system have been considered to be satisfied by different producers such as power only units, combined heating and power (CHP) plants, desalination only units, combined water and power (CWP) generation plants, in a way that minimum daily fuel cost is obtained for the studied system. In other words, each CHP producer not only participates in supplying the heat demand, but also generates a part of the electrical load. Moreover, the electrical load is economically dispatched between power only units, CHPs and CWPs. Hence, optimum operating point of each CHP unit should be determined by co-dispatching of mentioned plants. In other words, CHPs have been integrated with power only units and co-producers of water and power. Similarly, desalination only unit and co-generators of water and power have been considered for supplying the fresh water demand. Because of integration of CWP producers, CHPs units and power only plants in supplying the electrical demand, the operating point of the CHPs plants and power only units affect the best operating point of the CWP generation units. Hence, the operating point of the desalination only unit depends on the operating points of the CWP plants, which participate in drinking water production. The total fuel cost of the seawater desalination, heat treatment and electricity procurement processes is minimized as equation (1). Traditionally, the fuel cost of the coal and gas fired electricity generation processes (C_p) is a quadratic function versus the output electrical power of each thermal unit ($P_{g,t}$), which can be given by relation (2) [21,22]. Similarly, total fuel cost of CWP producers (C_{cwp}) and desalination only units (C_w) can be modelled as equations (3) and (4), respectively [23]. In the same manner, the fuel cost of boilers (C_h) and CHP plants (C_{chp}) can respectively be stated as two quadratic functions (5) and (6) versus output heating flow extracted from each boiler ($Q_{k,t}$), electricity ($P_{j,t}^{chp}$) and heat ($Q_{j,t}^{chp}$) productions of CHP units.

$$\text{Objective function} = C_p + C_{cwp} + C_w + C_h + C_{chp} \quad (1)$$

$$C_p = \sum_{g,t} (a_g P_{g,t}^2 + b_g P_{g,t} + c_g U_{g,t}); \quad \forall g \in \Omega_p, \forall t \in T \quad (2)$$

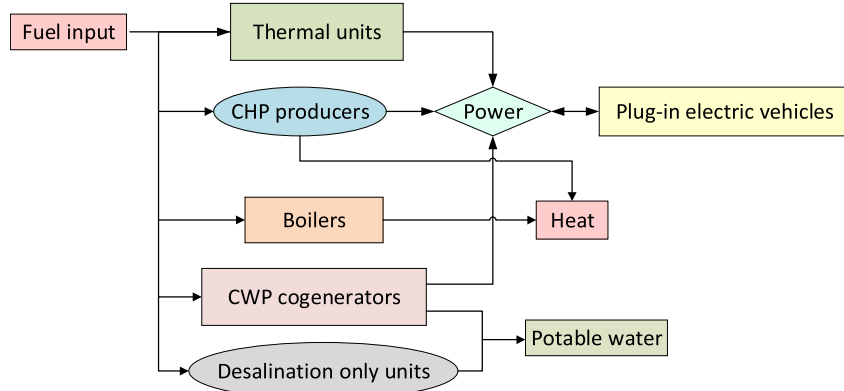


Fig. 1. The single line diagram of the heat-water-power hub network.

$$C_{cwp} = \sum_{c,t} (\alpha_c P_{c,t}^2 + \beta_c P_{c,t} W_{c,t} + \gamma_c W_{c,t}^2 + \zeta_c P_{c,t} + \varsigma_c W_{c,t} + \xi_c U_{c,t}); \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (3)$$

$$C_w = \sum_{w,t} (a_w W_{w,t}^2 + b_w W_{w,t} + c_w U_{w,t}); \quad \forall w \in \Omega_w, \forall t \in T \quad (4)$$

$$C_h = \sum_{k,t} (a_k Q_{k,t}^2 + b_k Q_{k,t} + c_k U_{k,t}); \quad \forall k \in \Omega_h, \forall t \in T \quad (5)$$

$$C_{chp} = \sum_{j,t} a_j^{chp} (P_{j,t}^{chp})^2 + b_j^{chp} P_{j,t}^{chp} + c_j^{chp} + d_j^{chp} (Q_{j,t}^{chp})^2 + e_j^{chp} Q_{j,t}^{chp} + f_j^{chp} P_{j,t}^{chp} Q_{j,t}^{chp}; \quad \forall j \in \Omega_{chp}, \forall t \in T \quad (6)$$

The objective cost function explained by (1) is minimized considering the following constraints [24,25]:

■ Electrical load-generation balance constraint

As stated by (7), the thermal unit and the co-producers should be able to supply the electrical demand of the test system at each hour [26].

$$\sum_{g \in \Omega_p} P_{g,t} + \sum_{c \in \Omega_{cwp}} P_{c,t} + \sum_{j \in \Omega_{chp}} P_{j,t}^{chp} = E_t; \quad \forall t = 1, 2, \dots, T \quad (7)$$

■ Water production by desalination unit and CWP plants

The output fresh water of the desalination plant and CWP units should be equal to the water consumption of the benchmark grid [23,27].

$$\sum_{c \in \Omega_{cwp}} W_{c,t} + \sum_{w \in \Omega_w} W_{w,t} = W_t; \quad \forall t = 1, 2, \dots, T \quad (8)$$

■ Heat supply-demand balance constraint

Similar to constraints (7) and (8), equation (9) expresses that CHP plants and heat only units participate in supplying the total

heating demand at time period t [28].

$$\sum_{j \in \Omega_{chp}} Q_{j,t}^{chp} + \sum_{k \in \Omega_h} Q_{k,t} = Q_t; \quad \forall t = 1, 2, \dots, T \quad (9)$$

■ Energy, heat and pure water production limits

The values of the electricity produced by the thermal power plants, CWP and CHP units are limited by upper and lower bounds as (10)–(12), respectively [29,30]. In addition, the feasible operating region of the CHP units is modelled in Fig. 2. It should be mentioned that the electrical power and heat generation of the CHP units depend on each other in a way that one output will impress other [31]. In other words, $P_{j,t}^{chp}$ and $Q_{j,t}^{chp}$ are non-separable as considered by constraints (12) and (13), respectively. Similarly, the volume of the pure water generation of the CWP units and desalination plants are limited in (14) and (15), respectively. In addition, the heating capacity of the boilers can be stated by constraint (16) [32]. The binary decision variables $U_{g,t}$, $U_{c,t}$, U_j^{chp} , $U_{k,t}$ and $U_{w,t}$ have been used to optimally make the on/off status decisions of the thermal generation units, co-producers, CHPs, boilers and desalination only units, respectively. In other words, all units have not been considered to be on at each time interval t . These binary decision variables have optimally been found by solving a mixed integer nonlinear programming problem aiming to minimize the

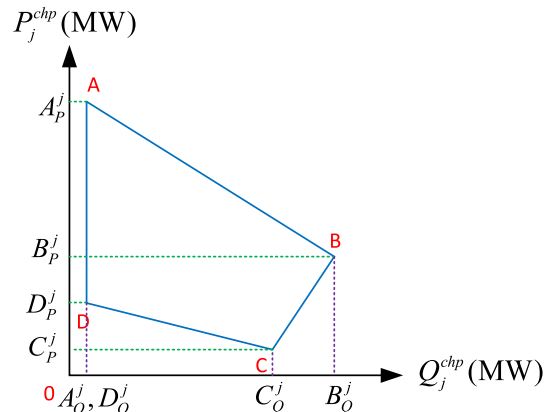


Fig. 2. The feasible operating region of the CHP unit j .

Table 1

Fuel cost factors and feasible operating regions of CHP producers.

Units	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	A_Q^j	A_P^j	B_Q^j	B_P^j	C_Q^j	C_P^j	D_Q^j	D_P^j
<i>j</i> = 1	0.0345	14	2540	0.03	4.2	0.031	0	247	180	215	104.8	81	0	99
<i>j</i> = 2	0.0435	13	1460	0.02	0.7	0.011	0	125	135	110	75	40	0	45

total operation cost of the water-heat-power generation system, while satisfying the potable water, heating and electrical demands of the benchmark grid. According to Fig. 2 and Table 1, when the idea is to produce only power without recovering the waste heat, the parameter A_Q^j will be zero and the value of the D_P^j should be not zero, which have been considered in Table 1. Similar to Refs. [11–13], this concept has been considered for co-producers of desalinated water and power as their power-water generation ratio limit, which has been formulated by constraint (18) and considered in optimization process.

$$P_g^{\min} \times U_{g,t} \leq P_{g,t} \leq P_g^{\max} \times U_{g,t}; \quad \forall g \in \Omega_p, \forall t \in T \quad (10)$$

$$P_c^{\min} \times U_{c,t} \leq P_{c,t} \leq P_c^{\max} \times U_{c,t}; \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (11)$$

$$\begin{aligned} P_{j, \text{chp}}^{\min} (Q_{j,t}^{\text{chp}}) \times U_{j,t}^{\text{chp}} &\leq P_{j,t}^{\text{chp}} \\ &\leq P_{j, \text{chp}}^{\max} (Q_{j,t}^{\text{chp}}) \times U_{j,t}^{\text{chp}}; \quad \forall j \in \Omega_{\text{chp}}, \forall t \in T \end{aligned} \quad (12)$$

$$\begin{aligned} Q_{j, \text{chp}}^{\min} (P_{j,t}^{\text{chp}}) \times U_{j,t}^{\text{chp}} &\leq Q_{j,t}^{\text{chp}} \\ &\leq Q_{j, \text{chp}}^{\max} (P_{j,t}^{\text{chp}}) \times U_{j,t}^{\text{chp}}; \quad \forall j \in \Omega_{\text{chp}}, \forall t \in T \end{aligned} \quad (13)$$

$$W_w^{\min} \times U_{w,t} \leq W_{w,t} \leq W_w^{\max} \times U_{w,t}; \quad \forall w \in \Omega_w, \forall t \in T \quad (14)$$

$$W_c^{\min} \times U_{c,t} \leq W_{c,t} \leq W_c^{\max} \times U_{c,t}; \quad \forall c \in \Omega_{cwp}, \forall t \in T \quad (15)$$

$$Q_k^{\min} \times U_{k,t} \leq Q_{k,t} \leq Q_k^{\max} \times U_{k,t}; \quad \forall k \in \Omega_h, \forall t \in T \quad (16)$$

■ Ramp rates of conventional thermal power plants

The conventional thermal power plants increase and decrease their output power based on ramp up and ramp down rates, as stated by (17) [33].

$$\begin{aligned} \text{Max} (P_g^{\min}, P_g^{t-1} - DR_g) &\leq P_{g,t} \\ &\leq \text{Min} (P_g^{\max}, P_g^{t-1} \\ &\quad + UR_g); \quad \forall g \in \Omega_p, \forall t \in T \end{aligned} \quad (17)$$

■ Power-water production rate

Similar to CHP units and constraints (12) and (13), a power-water ratio constant is considered for coproduction of power and water by unit *c* as formulated in (18) [27].

$$R_c^{\min} \leq \frac{P_{c,t}}{W_{c,t}} \leq R_c^{\max}; \quad \forall g \in \Omega_p, \forall t \in T \quad (18)$$

2.2. Aggregated plug-in electric vehicles

Electric drive vehicles offer some advantages such as zero-emission footprints, charging electricity during off-peak periods and discharging it for satisfying on-peak demand that is not available in conventional internal combustion engine vehicles. The contribution of the electric vehicle *i* at time *t* is modelled as follows: (a) Driving mode, (b) Charging mode, (c) Discharging mode, (d) None. Driving mode: The daily driving patterns of the PEVs are presented as the input parameters of the optimization problem and shown in Figs. 4–6. As stated in constraint (25), when the electrical energy requirement of the *i*th PEV at hour *t* is non-zero and it is driven ($E_{t,i}^{\text{PEV}} \neq 0$), this PEV cannot exchange the power with the local power system. Hence, the value of the power charge and discharge will be zero ($P_{t,i}^{\text{PEVch}}, P_{t,i}^{\text{PEVdis}} = 0$). If the binary decision variable $U_{t,i}^{\text{PEVch}}$ is equal to 1, the *i*th PEV will be connected to local power system and charged by electrical power $P_{t,i}^{\text{PEVch}}$ (grid-to-vehicle mode). If the binary variable $U_{t,i}^{\text{PEVdis}}$ is equal to 1, the *i*th PEV will be connected to local power system and discharged the electrical power $P_{t,i}^{\text{PEVdis}}$ (vehicle-to-grid mode). In other words, the connection status and the value of the power charge or discharge have not been shown in the daily driving patterns of the PEVs. The binary decision variables $U_{t,i}^{\text{PEVch}}$ and $U_{t,i}^{\text{PEVdis}}$ are used for determining when the PEVs are connected to the local power system, charged and discharged. In addition, the positive decision variables $P_{t,i}^{\text{PEVch}}$ and $P_{t,i}^{\text{PEVdis}}$ are utilized for finding the value of the charged and discharged power, respectively. At all hours that the energy consumption of the *i*th PEV is non-zero, the variables $P_{t,i}^{\text{PEVch}}$ and $P_{t,i}^{\text{PEVdis}}$ will be equal to zero as formulated by constraint (25). Hence, $U_{t,i}^{\text{PEVch}}$ and $U_{t,i}^{\text{PEVdis}}$ will be equal to zero as expected from constraints (20) and (21), respectively. Equation (24) imposes that SOC of battery pack *i* at hour *t* depends on its SOC at previous hour and value of charged power/discharged power/energy consumption at time *t*. Moreover, SOC of battery pack *i* at each time period should be less than its nominal capacity, as given by (22). Constraint (23) demonstrates that *i*th PEV can be charged or discharged or none at each hour *t*.

$$\text{SOC}_{t_0}^{\text{PEVi}} = \text{SOC}_0^{\text{PEVi}}; \quad \forall i \quad (19)$$

$$P_{t,i}^{\text{PEVch}} \times U_{t,i}^{\text{PEVch}} \leq P_{t,i}^{\text{PEVch}} \leq P_{t,i}^{\text{PEVch}} \times U_{t,i}^{\text{PEVch}} \quad (20)$$

$$P_{t,i}^{\text{PEVdis}} \times U_{t,i}^{\text{PEVdis}} \leq P_{t,i}^{\text{PEVdis}} \leq P_{t,i}^{\text{PEVdis}} \times U_{t,i}^{\text{PEVdis}} \quad (21)$$

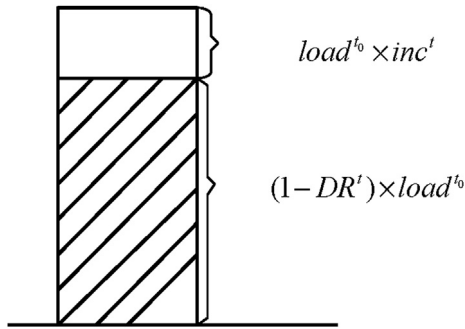


Fig. 3. The model of the DRPs.

$$SOC_{PEV_i}^{min} \leq SOC_t^{PEV_i} \leq SOC_{PEV_i}^{max} ; \quad \forall t, \forall i \quad (22)$$

$$U_{t,i}^{PEV_{ch}} + U_{t,i}^{PEV_{dis}} \leq 1 ; \quad \forall t, \forall i \quad (23)$$

$$SOC_t^{PEV_i} = SOC_{t-1}^{PEV_i} + \chi \times P_{t,i}^{PEV_{ch}} - E_{t,i}^{PEV} - \frac{P_{t,i}^{PEV_{dis}}}{\eta} ; \quad \forall t, \forall i \quad (24)$$

$$P_{t,i}^{PEV_{ch}}, P_{t,i}^{PEV_{dis}} = 0 ; \left\{ \forall t, \forall i \mid E_{t,i}^{PEV} \neq 0 \right\} \quad (25)$$

2.3. Demand response programs

In this research, demand response programs (DRPs) are applied on water, heat and power demands. As illustrated in Fig. 3 and according to equation (26), $load^{t_0} (1 - DR^t)$ is related to non-flexible loads that don't participate in DRPs. The term $load^{t_0} \times inc^t$ is related to the demand reduction of the flexible loads. At each hour t , we have $DR^t \leq DR_{max}$ as stated by equation (28). Hence, the optimization problem finds the best value of DR^t and then $load^{t_0} (1 - DR^t)$ for each hour in a way that not only sum of load reductions should be equal to sum of load increments as shown in equation (27), but also the percentage of load increase (inc^t) should be smaller than inc_{max} as formulated in constraint (29) and the objective cost function is minimized as low as possible. If there is no participation in DRPs, $DR^t = 0$ and $load^{t_0} \times DR^t = 0$ and the load has no change at hour t . Some of electricity consumers have been registered in DRPs and called flexible or shiftable loads. In this study, only 20% of base water/heat/power load is participated in

DRPs and $DR_{max} = 0.2$. Moreover, 80% of each demand is considered as nonshiftable load. In addition, the percentage of the demand reduction is considered to be less than the 20% as formulated by inequality constraint (28).

$$load^t = load^{t_0} (1 - DR^t) + load^{t_0} \times inc^t \quad (26)$$

$$\sum_{t=1}^T (load^{t_0} \times inc^t) = \sum_{t=1}^T (load^{t_0} \times DR^t) \quad (27)$$

$$DR^t \leq DR_{max} \quad (28)$$

$$inc^t \leq inc_{max} \quad (29)$$

3. Case study

In this study, a sample water-heat-power production system composed of two CHP units, two heating only processes, three CWP producers, one water desalination plant, and four thermal generating units [27] is comprehensively studied.

3.1. Inputs

The operational specifications of the mentioned units are provided in Tables 1–5. The presented approach is modelled as a mixed integer nonlinear program (MINLP) and solved using the branch-and-reduce optimization navigator (BARON) [34] of the general algebraic mathematical modelling system (GAMS) [35]. In this research, four cases are studied to reveal that integration of seawater desalination processes, boilers, CWP and CHP plants, and power generation units is a cost-effective strategy in comparison with that of their separable operation especially while incorporating PEVs and real-time DRPs.

Case 1. Short-term optimal coordination of water-heat-power nexus without PEVs and DRPs;

Case 2. Day-ahead co-scheduling of seawater desalination, heat

Table 4
Constant fuel cost factors of desalination only plant.

Unit	a_w	b_w	c_w	W_w^{min}	W_w^{max}
$w = 1$	0.00182	−0.374	7.374	0	250

Table 5
Operational characteristics of conventional thermal power plants.

Units	a_g	b_g	c_g	p_g^{min}	p_g^{max}
$g = 1$	0.0002069	−0.1483	57.11	0	500
$g = 2$	0.0003232	−0.1854	57.11	0	400
$g = 3$	0.001065	−0.6026	126.8	0	400
$g = 4$	0.0004222	−0.2119	57.11	0	350

Table 2
Fuel cost coefficients of boilers.

Units	a	b	c	Q_k^{min}	Q_k^{max}
$k = 1$	4.05	10.55	104.26	60	200
$k = 2$	3.99	9.21	107.21	70	270

Table 3
Fuel cost indices of CWP units.

Units	α_c	β_c	γ_c	ζ_c	ς_c	ξ_c	p_c^{min}	p_c^{max}	W_c^{min}	W_c^{max}	R_c^{min}	R_c^{max}
$c = 1$	0.0004433	0.003546	0.007093	−1.106	−4.426	737.4	160	800	30	200	4	9
$c = 2$	0.0007881	0.006305	0.01261	−1.475	−5.901	737.4	120	600	23	150	4	9
$c = 3$	0.001773	0.01419	0.02837	−2.213	−8.851	737.4	80	400	15	100	4	9

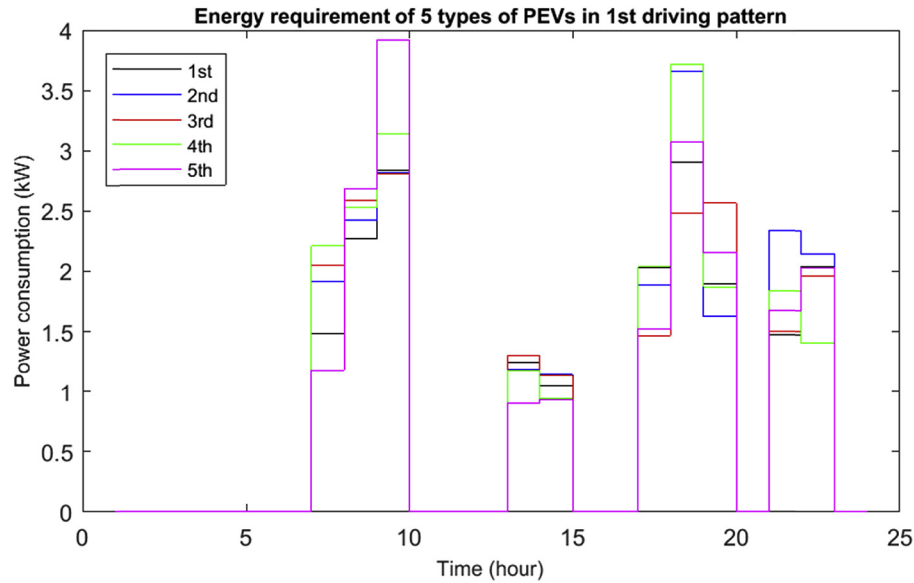


Fig. 4. First daily driving pattern for five PEVs.

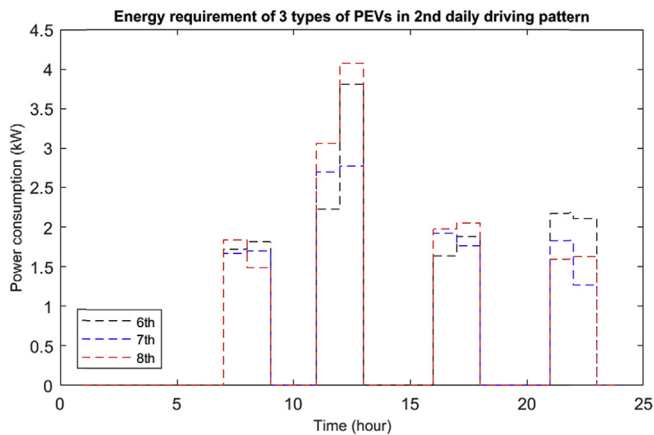


Fig. 5. Second daily driving pattern for three PEVs.

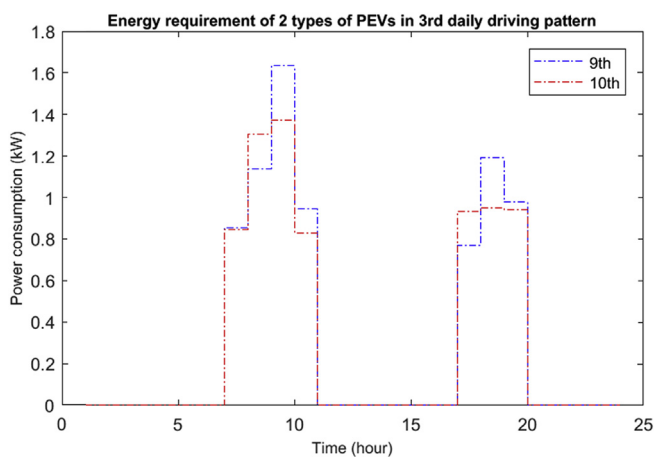


Fig. 6. Third daily driving pattern for two PEVs.

Case 3. With day-ahead co-dispatching of PEVs in water-heat-power generation units;

Case 4. With optimal participation of PEVs and implementation of DRPs on water-heat-power nexus.

Moreover, a set of three different driving patterns for 10 new electric drive vehicles have been considered as Figs. 4–6. Number and daily driving patterns of PEVs have been considered similar to those of provided in Ref. [36]. As reported in Table 6, 1000 PEVs with 25 kW capacity are optimally participated in plant level energy procurement process. In other words, the total power generation capacity of the PEVs are considered to be equal to 25 MW, which is a significant value for participation in electricity generation facility. Other parameters related to charge and discharge decisions of PEVs used in cases 3 and 4 are presented in Table 6. In addition, the hourly variations of the fresh water, heat and the electrical demands during a 24-h time horizon are depicted in Fig. 7. It should be noted that these loads are associated with different consumers such as residential, industrial, commercial, agricultural sectors located in the studied region. In other words, the sum of the electricity/heat/water consumption of these customers has been considered as the electrical/heat/water demand of the test system.

3.2. Electricity generated by power only units, CHP producers and CWP plants

As shown in Tables 1, 3 and 5, the maximum electrical power of CHP producers, CWP plants and the conventional thermal units are equal to 247 MW (CHP unit 1), 125 MW (CHP unit 2), 800 MW (CWP unit 1), 600 MW (CWP unit 2), 400 MW (CWP unit 3), 500 MW (power only unit 1), 400 MW (power only units 2 and 3), 350 MW

Table 6

Technical specifications related to charge and discharge modes of pure electric vehicles.

$SOC_{PEV_i}^{min}$, $SOC_{PEV_i}^{max}$ [kW]	χ, η	$p_{PEV_{ch}}^{min}$, $p_{PEV_{dis}}^{min}$ [kW]	$p_{PEV_{ch}}^{max}$, $p_{PEV_{dis}}^{max}$ [kW]	Number of PEVs
0, 25	1	0	25	1000

and power generation plants with implementation of DRPs on potable water, heat and electricity demands;

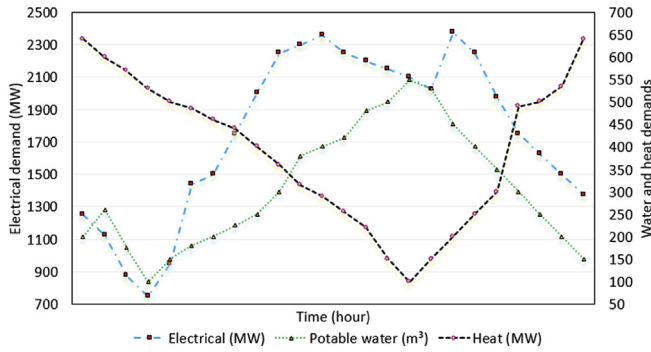


Fig. 7. Total desalinated water, heat and electricity loads over a 24-h study horizon.

(power only unit 4). According to Fig. 7, the maximum electrical, heat and desalinated water demands are equal to 2375 MW, 640 MW and 550 m³ which occur at $t = 18$, $t = 24$ and $t = 16$, respectively. As mentioned, the electricity generation capacities of the conventional power plants, CHP units, and CWP producers are equal to 1650 MW, 372 MW and 1800 MW, respectively. Hence, it is infeasible to satisfy the on-peak electrical demand either by their separate operations or by the cooperation of power only units and CHP plants. In other words, CWP and thermal generators should be co-scheduled to supply 2375 MW electrical demand. Moreover, the water only unit produces a maximum volume of the fresh water up to 250 m³ which cannot satisfy the desalinated water demand at $t = 2$ and $t = 9$ to 22. Therefore, CWP producers and water only units are co-dispatched. As shown in Table 2, it is obvious that the heat only units and CHP plants should also be integrated for supplying 640 MW on-peak heating demand at $t = 1$ and $t = 24$. As mentioned load-generation balance criteria, a water-heat-power nexus is optimally scheduled by solving the MINLP problem (1)–(29). The hourly generation schedules of thermal units, CHP plants and CWP producers in four cases without and with application of DRPs and participation of PEVs have optimally been found as shown by Figs. 8–10, respectively. As obvious from these figures, their optimum operating point change after implementation of real-time demand-side management strategy and participation of electric drive vehicles in a way that total fuel cost of power only units (C_p) reduces from \$1029 (Case 1: without DRPs and PEVs) to \$839,131 (Case 2: with DRPs), \$1023 (Case 3: with PEVs) and \$924 (Case 4: with DRPs and PEVs), as shown in Table 7.

3.3. Heat extracted from boilers and CHP units

According to Table 2, it is obvious that maximum heating capacity of heat only units 1 and 2 are equal to 200 MW and 270 MW, respectively. As illustrated in Fig. 7, the maximum heating demand of the test system is equal to 640 MW. Hence, boilers cannot supply the heat consumers without participation of CHP units at hours $t = 1$ and $t = 24$. As expressed in subsection 3.2, CHP producers should also be optimally co-scheduled with power only units and CWP plants to satisfy on-peak electrical load. Meanwhile, economic operating point (power generation and heat extraction values) of CHP units depend on electrical power output of CWP plants and thermal generation stations. Similarly, electricity productions of CWP units and then power only unit changes by economic operation of desalination plant. Optimum hourly schedules of CHP units and boilers for supplying total heating demand are obtained as Figs. 11 and 12, respectively. According to Table 7, total fuel cost of heat only units (C_h) reduces from \$1,836,635 (Case 1: without DRPs and PEVs) to \$1,153,000 (Case 2: with DRPs), \$1,698,121 (Case 3:

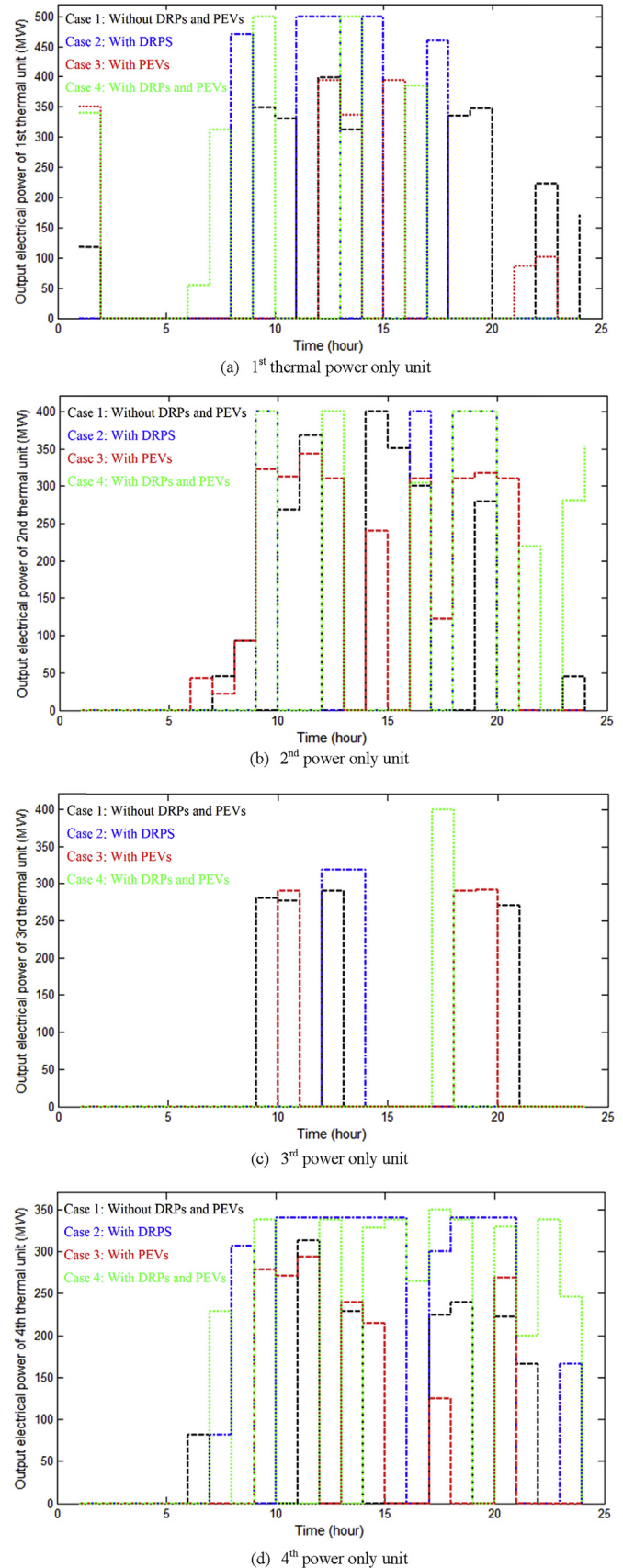


Fig. 8. Output electrical power of conventional thermal generating units.

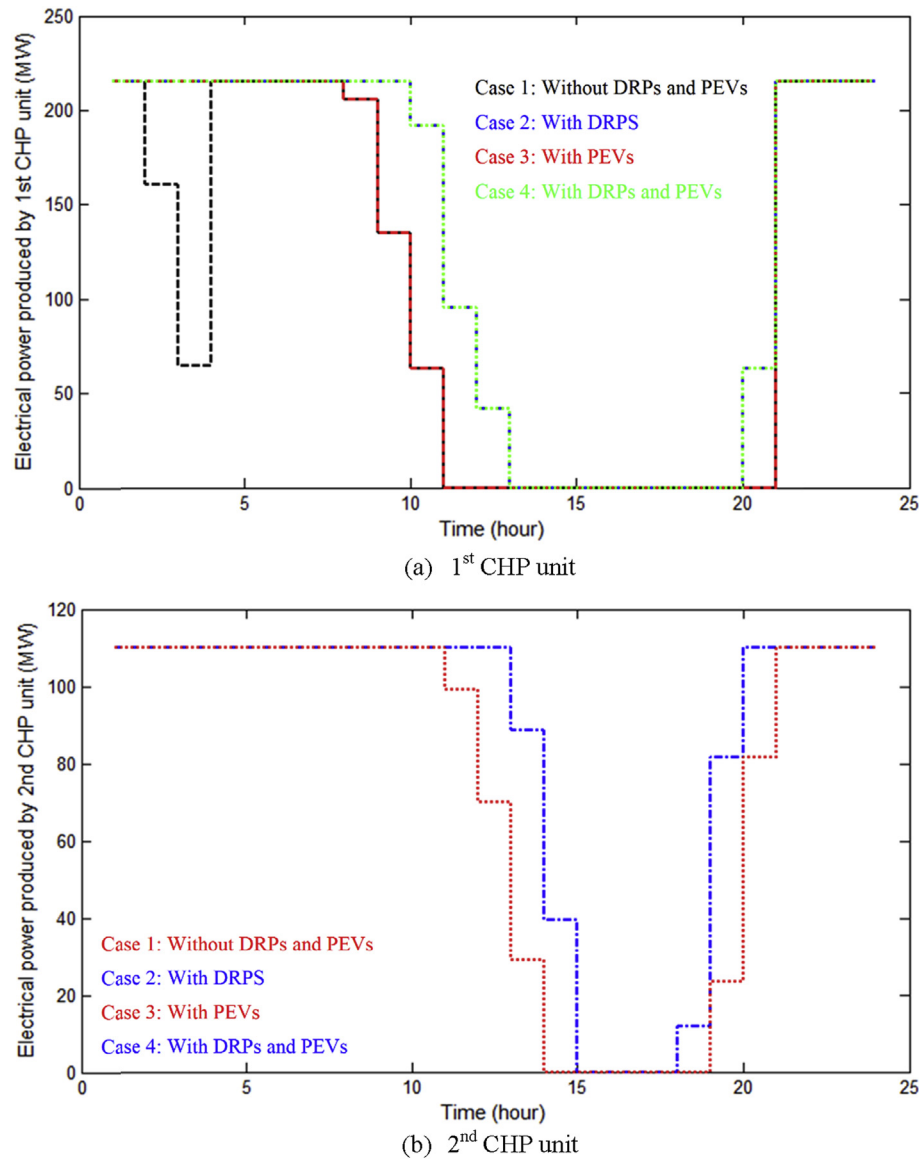


Fig. 9. Electrical power produced by CHP generation companies.

with PEVs) and \$1,152,979 (Case 4: with DRPs and PEVs). Fig. 12 and Table 7 demonstrate that if DRPs and PEVs are incorporated in heat-water-power nexus, CHP generation companies will participate more than thermal power plants and heat only units in supplying total electricity and heat demands causing an increase in their total fuel cost (C_{chp}).

3.4. Potable water produced by desalination unit and CWP plants

The potable water generation capacity of the desalination only unit is equal to 250 m^3 , which cannot satisfy the fresh water requirement of the benchmark system at hours $t = 2$ and $t = 9$ to 22 . Hence, it is suggested that CWP cogeneration facilities and desalination only unit are optimally co-dispatched. According to electrical load-generation balance constraint explained in subsection 3.2, CWP units should also be co-scheduled with thermal power plants and CHP producers for satisfying on-peak electricity load. Therefore, optimal operating points (electricity and water productions) of CWP units depend on electrical power

output of thermal power plants and CHPs. By solving the optimization problem (1)–(29), the best operating points of desalination only unit and CWP producers for production of fresh water are obtained as Figs. 13 and 14, respectively. It is found that total fuel cost of desalination only unit (C_w) is equal to zero in four cases because of negative sign of b_w which is reported in Table 4. Meanwhile, total fuel cost of CWP plants (C_{cwp}) reduces from \$26,853 (Case 1: without DRPs and PEVs) to \$23,195 (Case 2: with DRPs), \$26,696 (Case 3: with PEVs) and \$23,457 (Case 4: with DRPs and PEVs).

3.5. Optimal participation of PEVs and DRPs in power-heat-water hub network

As mentioned in subsection 2.3, PEVs are an emerging technology with significant potential for reduction of gasoline consumptions and carbon dioxide emissions. In addition, incorporating a large number of PEVs with proper control or optimal charge and discharge decisions (time and value of power charge or discharge)

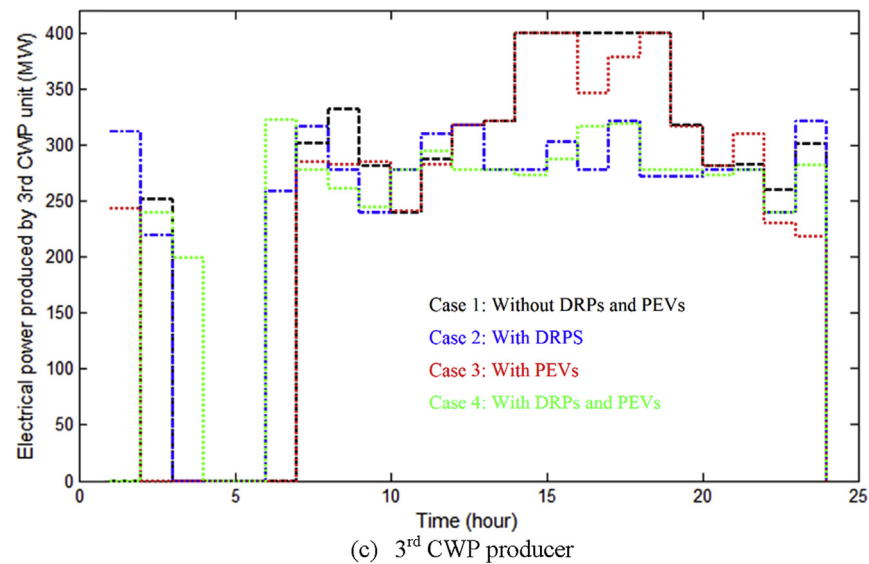
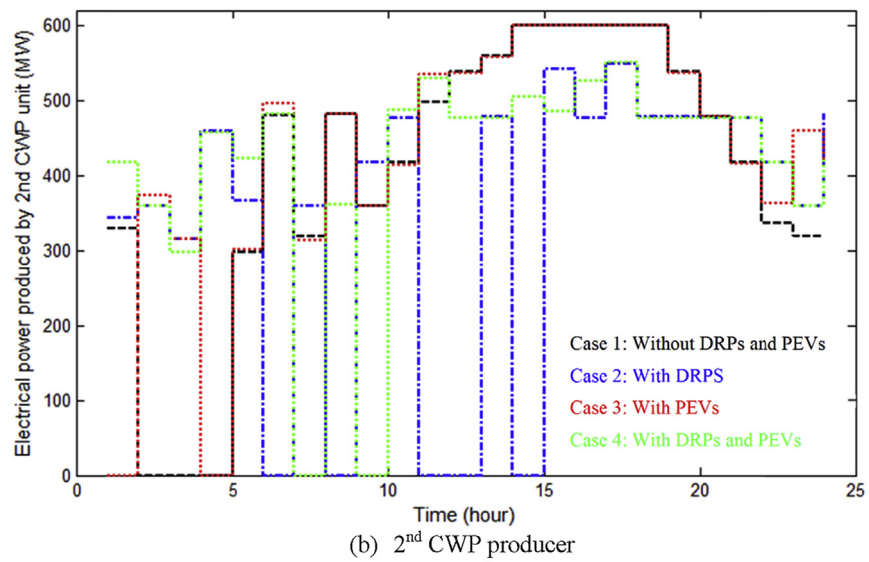
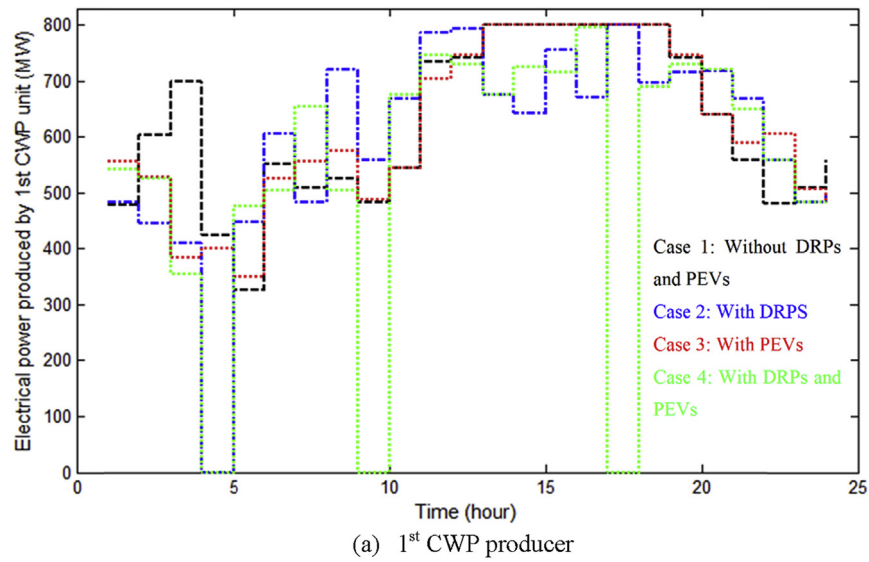


Fig. 10. Electrical power generated by CWP units.

Table 7
Economic comparison between four cases.

Case study	C_p	C_{cwp}	C_{chp}	C_h	Objective function [\\$]
1	1029	26,853	229,098	1,836,635	2,093,616
2	839	23,195	256,681	1,153,000	1,433,716
3	1023	26,696	237,056	1,698,121	1,962,897
4	924	23,457	256,681	1,152,979	1,434,043

could bring an opportunity of large storage and flexibility for energy systems. Hence, times and values of charge (grid-to-vehicle) and discharge (vehicle-to-grid) modes for each type of PEV are optimally scheduled while considering this fact that they can be charged and discharged only in plugged or non-driving times and cannot be charged and discharge simultaneously. In addition, by participating in DRPs, customers are responsible for changing their energy consumption pattern to reduce their expenses and daily operational cost of trigeneration system. In cases 2 and 4, optimal

operation of this system is investigated after application of DRPs in order to transfer some part of each load from the expensive (peak) periods to the cheaper (off-peak) time intervals, which leads to load flattening and cost reduction. Optimal charge and discharge decisions of some electric vehicles with different daily driving patterns are shown in Fig. 15. In this figure, the negative part of the vertical column refers to the charged power during grid to vehicle operating mode and the positive one indicates the discharged power over the vehicle to grid mode. By implementation of real-time DRPs with $DR_{max} = inc_{max} = 20\%$, maximum 20% of electrical, fresh water and heating loads can be shifted from high-price hours to low-cost periods as shown in Fig. 16. In other words, if their hourly patterns changes according to Fig. 16, total fuel cost of water-heat-power nexus (Objective function) reduces from \$2,093,616 (Case 1: without DRPs and PEVs) and \$1,962,897 (Case 3: without DRPs) to \$1,433,716 (Case 2: with DRPs) and \$1,434,043 (Case 4: with DRPs and PEVs), as reported in Table 7. As mentioned, 1000 PEVs with 25 kW capacity are optimally participated in plant level

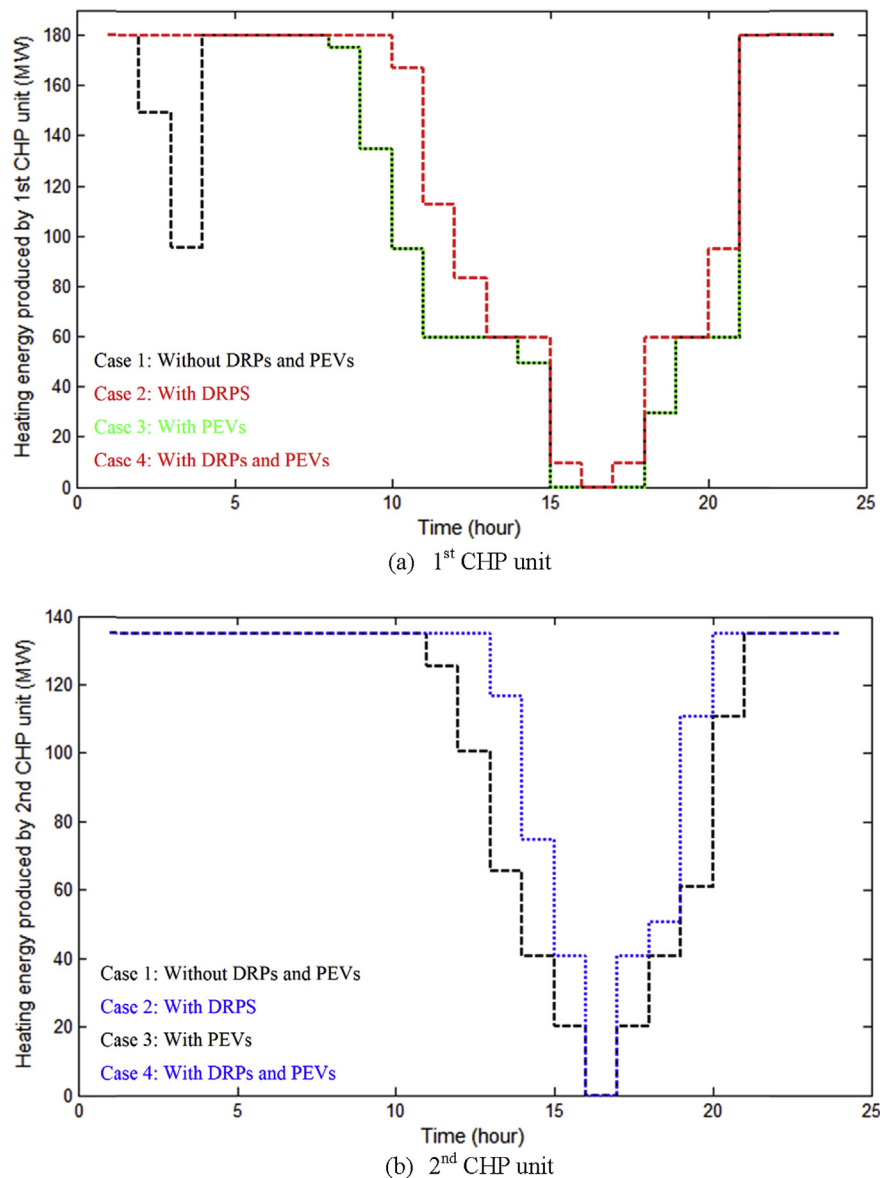


Fig. 11. Heating energy exhausted from CHP plants.

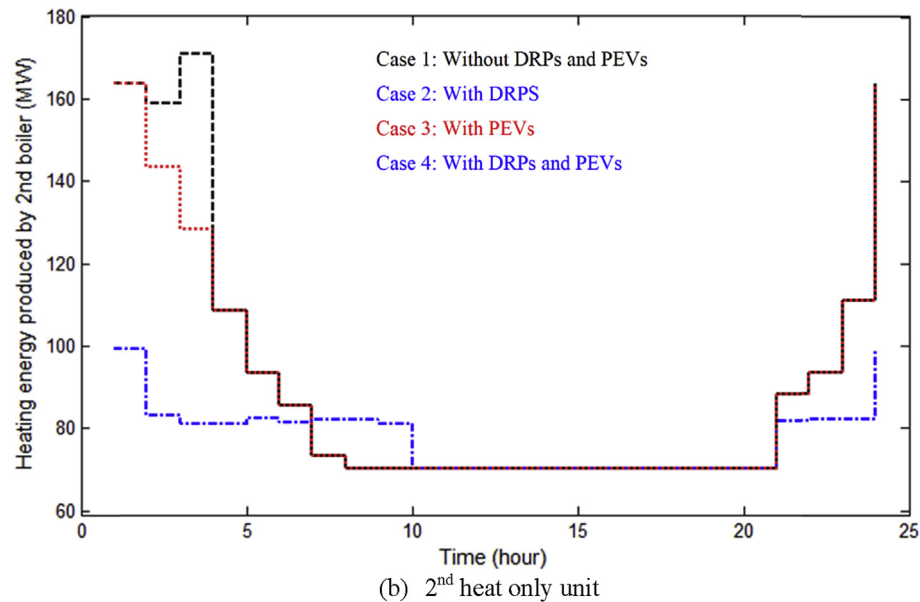
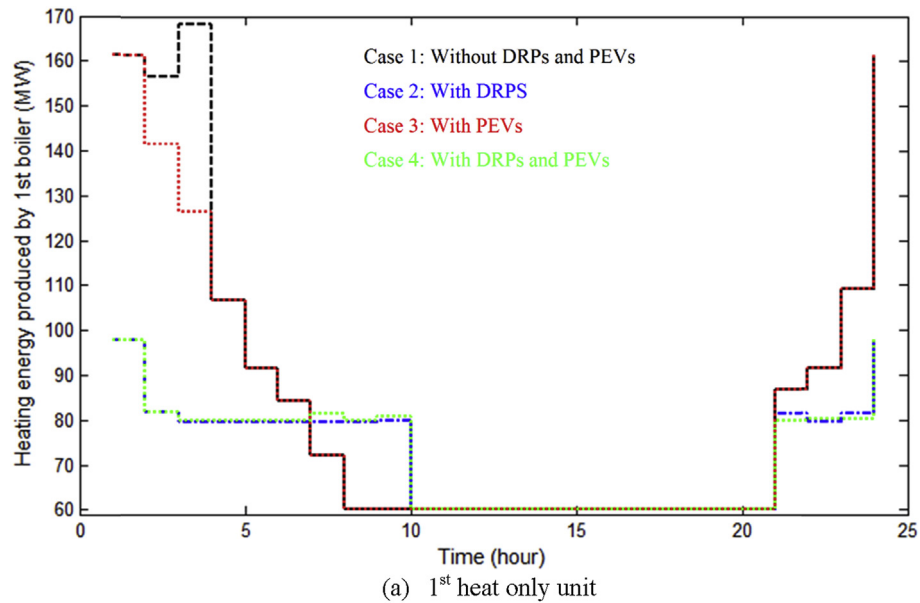


Fig. 12. Heating energy generated by boilers.

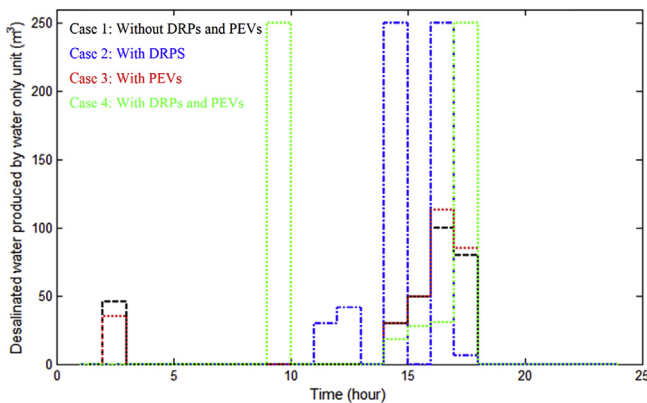


Fig. 13. Potable water produced by desalination process.

energy procurement process. In other words, the total power generation capacity of the PEVs are considered to be equal to 25 MW. As revealed in Table 7, the objective function of the water-heat-power nexus has been reduced from \$2,093,616 (case study 1: without participation of PEVs and DRPs in power generation process) to \$1,962,897 (case study 3: with incorporation of PEVs in energy procurement strategy). As demonstrated in this Table, optimization of charge and discharge patterns of pure electric vehicles with respect to their daily driving patterns not only causes a significant reduction in CO₂ footprints, but also leads to \$130,719 cost saving. For example, daily driving pattern of 4th type of PEVs is shown in Fig. 4. It is obvious from this figure that 4th type of PEVs, which is shown in green, is driven at hours 7–9, 13, 14, 17–19, 21 and 22. Hence, it cannot be connected to local power system at these hours. In other words, charge and discharge modes of 4th type of PEVs don't occur at mentioned times. At other hours, they

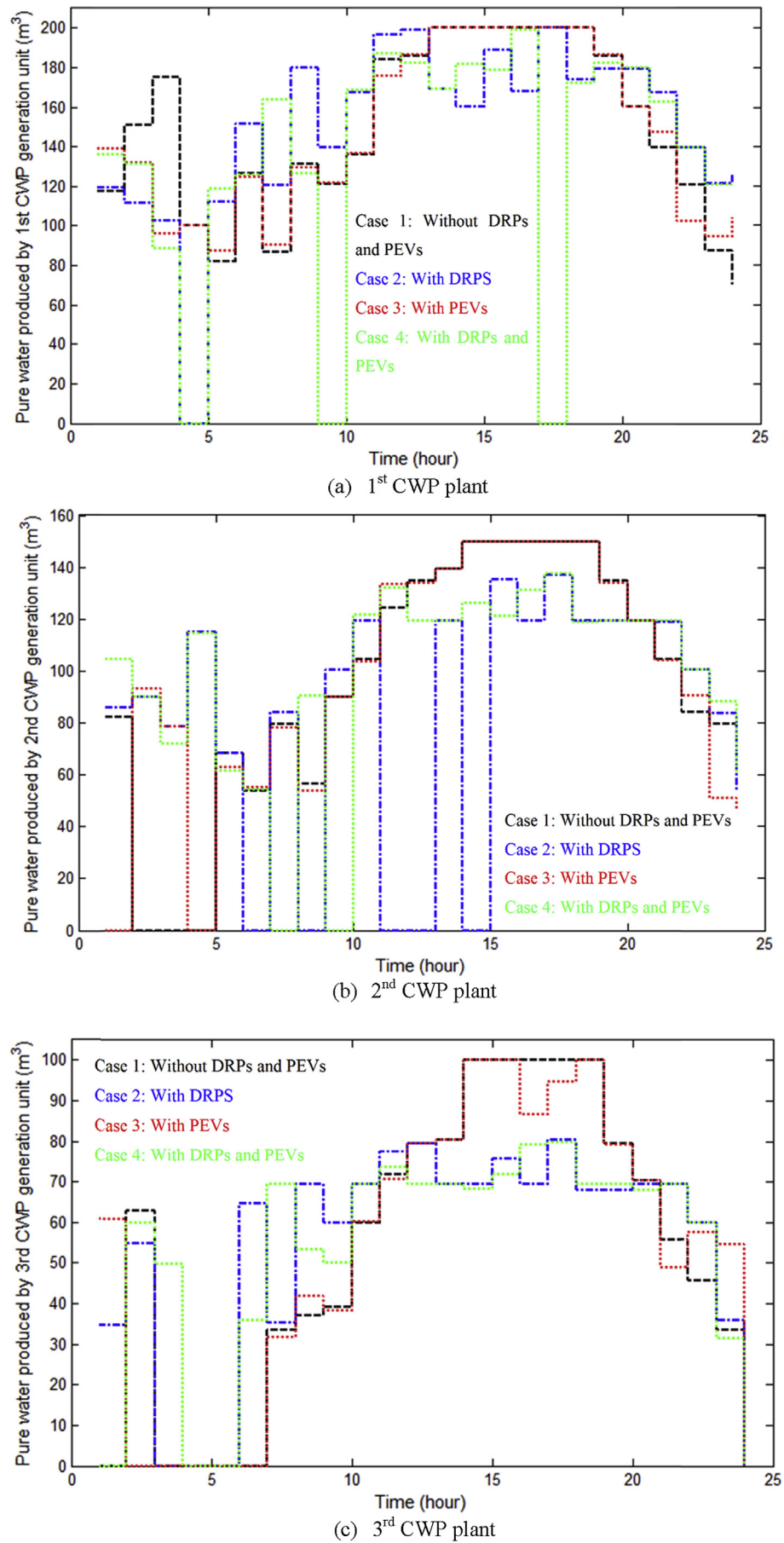


Fig. 14. Output fresh water of CWP producers.

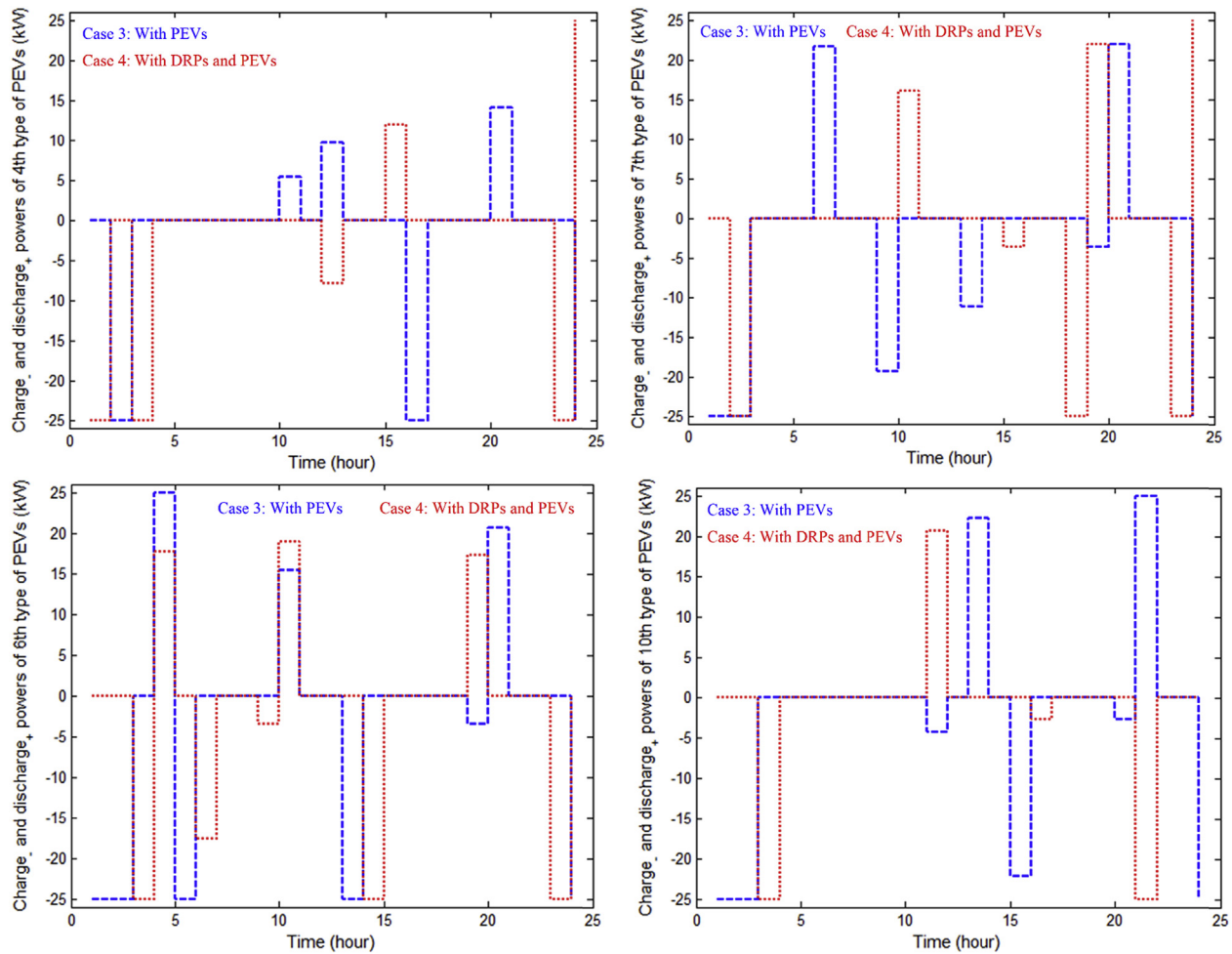


Fig. 15. Optimal charge and discharge decisions of 4th, 6th, 7th and 10th types of PEVs in two Cases.

can be charged or discharged or none as expected from constraint (23). As obtained from optimization process and illustrated in Fig. 15, it is found that 4th type of PEVs is discharged at hour 12 and “case 3: with PEVs” (blue profile in Fig. 15). In the case 3, it is charged at hour 12 and “Case 4: with PEVs and DRPs” (red profile in Fig. 15). In summary, it is revealed that for a \$130,719 cost saving in test water-heat-power generation system, not only there is no need to build a 25 MW power plant; but also 1000 PEVs participate in transmission sector according to their daily driving patterns, which are shown in Figs. 4–6. It may seem that the number of PEVs is low for a plant level energy production strategy, but their incorporation in water-heat-power generation system results in \$130,719 (6%) cost saving in a plant level multi-generation system. In comparison with Case 1, simultaneous incorporation of demand response programs and aggregated plug-in electric vehicles into water-heat-power trigeneration system (in case 4) leads to 31% cost saving. On a Lenovo with 2.10 GHz CPU, 4 GB RAM, GAMS optimization software is able to find the best operating point of the trigeneration system in less calculation time and lower computational burden than MATLAB software and evolutionary search algorithms. There is no need to apply a search algorithm such as particle swarm optimization (PSO), imperialistic competitive algorithm (ICA), genetic algorithm (GA), and etc. on MINLP problem (1)–(29). It is solved by BARON tool of GAMS software at 1.047 s. In addition, it is able to find the global optimal solution in the optimization process.

But, the other evolutionary search approaches find the local optimal solutions and their performance depends on the number of iterations.

4. Conclusion

This paper proposed a comprehensive model for optimization of water-heat-power trigeneration system over a 24-h study horizon. Moreover, implementation of time-amount based demand response programming strategy on desalinated water, heat and electricity demands of a benchmark tri-generation grid was investigated from an economic viewpoint. In addition, optimal charge and discharge decisions of aggregated plug-in electric vehicles is determined over at each operating time interval for minimizing the fuel cost of the presented facility, while participating in transportation sector. Numerical result revealed that the total electrical demand of this case study can be supplied neither by power only units nor by cooperation of thermal units and cogeneration facilities. Similar analysis can be considered for satisfying total heating demand by boilers and CHP plants. Hence, optimal co-scheduling of heat, water and electricity networks was developed. In addition, it is found that application of real-time demand response programs and incorporation of plug-in electric vehicles in seawater desalination, heat treatment and electricity generation processes, resulted in \$659,573 lower fuel cost.

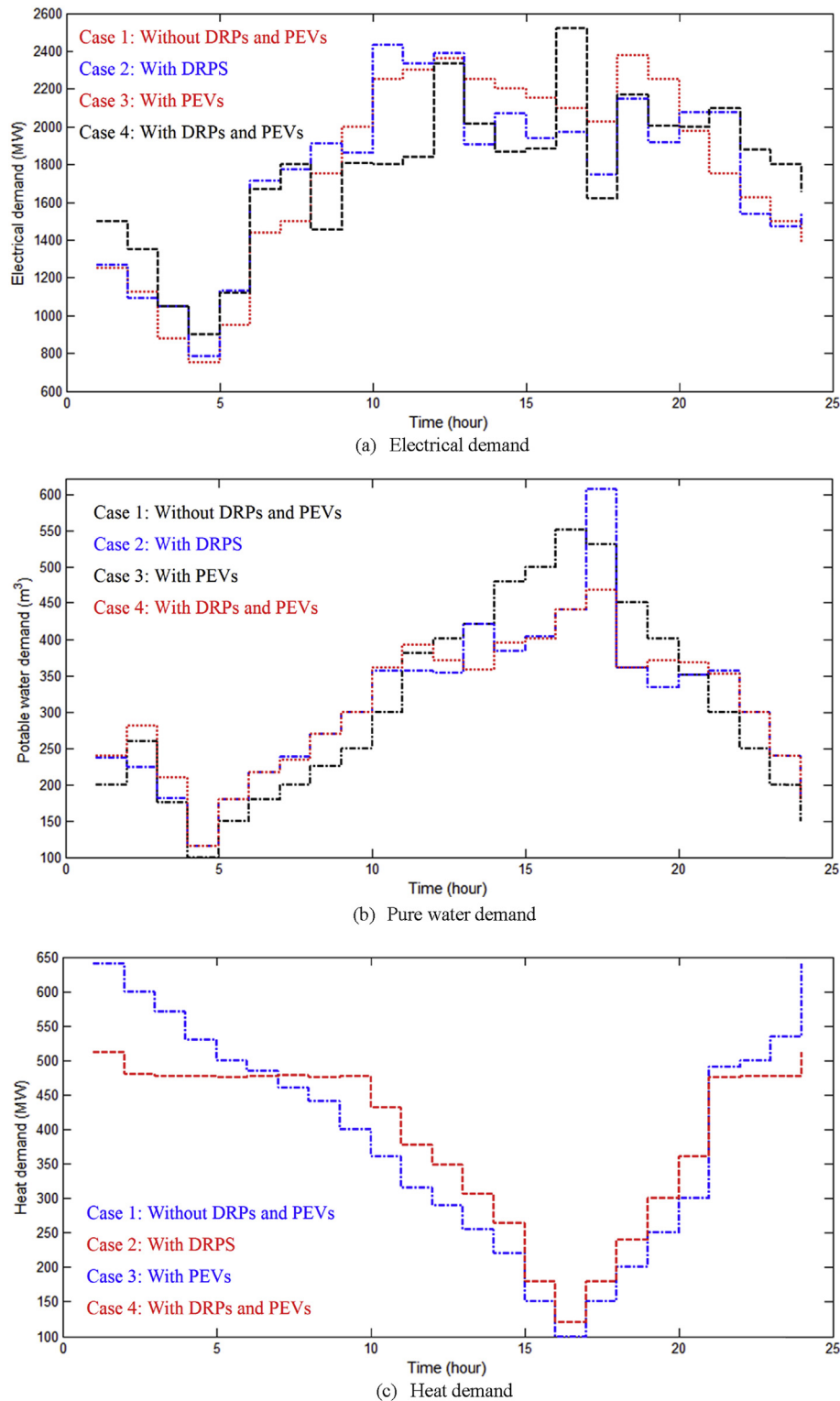


Fig. 16. Electrical, fresh water and heating loads after implementation of real-time DRPs.

References

- [1] Zhang X, Vesselinov VV. Integrated modeling approach for optimal management of water, energy and food security nexus. *Adv Water Resour* 2017;101: 1–10.
- [2] Administration USEI. International energy outlook 2016 with projections to 2040. 2016. p. 1–276.
- [3] Esmaili M, Shayanfar HA, Amjady N. Congestion management enhancing transient stability of power systems. *Appl Energy* 2010;87(3):971–81.
- [4] Sullivan MJ, Vardell T, Johnson M. Power interruption costs to industrial and commercial consumers of electricity. *IEEE Trans Ind Appl* 1997;33(6): 1448–58.
- [5] LaCommare K, Eto J. Cost of power interruptions to electricity consumers in the United States (US). *Energy* 2006;31(12):1845–55.
- [6] Ghazvini MAF, Faria P, Ramos S, Morais H, Vale Z. Incentive-based demand response programs designed by asset-light retail electricity providers for the

- day-ahead market. *Energy* 2015;82:786–99.
- [7] Nunna KHSVS, Doolla S. Responsive end-user-based demand side management in multimicrogrid environment. *IEEE Trans Ind Inf* 2014;10(2):1262–72.
 - [8] Park L, Jang Y, Cho S, Kim J. Residential demand response for renewable energy resources in smart grid systems. *IEEE Trans Ind Inf Dec.* 2017;13(6):3165–73.
 - [9] Nosratabadi SM, Hooshmand R-A, Gholipour E. Stochastic profit-based scheduling of industrial virtual power plant using the best demand response strategy. *Appl Energy* 2016;164(Supplement C):590–606.
 - [10] Vahid-Ghavidel M, Mahmoudi N, Mohammadi-ivatloo B. Self-scheduling of demand response aggregators in short-term markets based on information gap decision theory. *IEEE Trans Smart Grid* March 2019;10(2):2115–26.
 - [11] Hao H, Corbin CD, Kalsi K, Pratt RG. Transactive control of commercial buildings for demand response. *IEEE Trans Power Syst* 2017;32(1):774–83.
 - [12] Jabari F, Sharifian MBB, Mohammadi-ivatloo B. Optimized energy management of electric drive vehicles in a green residential building. *Majlesi J Electr Eng* 2017;11(4).
 - [13] Yang J, Chen B. Energy–water nexus of wind power generation systems. *Appl Energy* 2016;169(Supplement C):1–13.
 - [14] Siddiqi A, Kajenthira A, Anadón LD. Bridging decision networks for integrated water and energy planning. *Energy Strat Rev* 2013;2(1):46–58.
 - [15] Duan C, Chen B. Energy–water nexus of international energy trade of China. *Appl Energy* 2017;194:725–34.
 - [16] Zhang X, Vesselinov VV. Energy-water nexus: balancing the tradeoffs between two-level decision makers. *Appl Energy* 2016;183:77–87.
 - [17] Wa'el AH, Memon FA, Savic DA. A risk-based assessment of the household water-energy-food nexus under the impact of seasonal variability. *J Clean Prod* 2018;171:1275–89.
 - [18] Gjorgiev B, Sansavini G. Water-energy nexus: impact on electrical energy conversion and mitigation by smart water resources management. *Energy Convers Manag* 2017;148:1114–26.
 - [19] Gjorgiev B, Sansavini G. Electrical power generation under policy constrained water-energy nexus. *Appl Energy* 15 January 2018;210:568–79.
 - [20] Fuentes-Cortés LF, Ponce-Ortega JM. Optimal design of energy and water supply systems for low-income communities involving multiple-objectives. *Energy Convers Manag* 2017;151:43–52.
 - [21] Ma H, Yang Z, You P, Fei M. Multi-objective biogeography-based optimization for dynamic economic emission load dispatch considering plug-in electric vehicles charging. *Energy* 15 September 2017;135:101–11.
 - [22] Zou D, Li S, Li Z, Kong X. A new global particle swarm optimization for the economic emission dispatch with or without transmission losses. *Energy Convers Manag* 2017;139:45–70.
 - [23] Santhosh A, Farid AM, Youcef-Toumi K. Real-time economic dispatch for the supply side of the energy-water nexus. *Appl Energy* 2014;122:42–52.
 - [24] Shaabani Y a, Seifi AR, Kouhanjani MJ. Stochastic multi-objective optimization of combined heat and power economic/emission dispatch. *Energy* 2017;141:1892–904.
 - [25] Shi B, Yan L-X, Wu W. Multi-objective optimization for combined heat and power economic dispatch with power transmission loss and emission reduction. *Energy* 2013;56:135–43.
 - [26] Narang N, Sharma E, Dhillon JS. Combined heat and power economic dispatch using integrated civilized swarm optimization and Powell's pattern search method. *Appl Soft Comput* 2017;52:190–202.
 - [27] Soroudi A. Energy system integration. *Power system optimization modeling in GAMS*. Springer; 2017. p. 265–92.
 - [28] Li Y, Wang J, Zhao D, Li G, Chen C. A two-stage approach for combined heat and power economic emission dispatch: combining multi-objective optimization with integrated decision making. *Energy* 2018;162:237–54.
 - [29] Mohammadi-ivatloo B, Moradi-Dalvand M, Rabiee A. Combined heat and power economic dispatch problem solution using particle swarm optimization with time varying acceleration coefficients. *Electr Power Syst Res* 2013;95:9–18.
 - [30] Santhosh A, Farid AM, Youcef-Toumi K. The impact of storage facility capacity and ramping capabilities on the supply side economic dispatch of the energy–water nexus. *Energy* 2014;66:363–77.
 - [31] Beigvand SD, Abdi H, La Scala M. Combined heat and power economic dispatch problem using gravitational search algorithm. *Electr Power Syst Res* 2016;133:160–72.
 - [32] Bahmani-Firouzi B, Farjah E, Seifi A. A new algorithm for combined heat and power dynamic economic dispatch considering valve-point effects. *Energy* 2013;52:320–32.
 - [33] Nguyen TT, Vo DN, Dinh BH. Cuckoo search algorithm for combined heat and power economic dispatch. *Int J Electr Power Energy Syst* 2016;81:204–14.
 - [34] <https://www.gams.com/docs/solvers/baron/>.
 - [35] <https://www.gams.com/>.
 - [36] Alipour M, Mohammadi-ivatloo B, Moradi-Dalvand M, Zare K. Stochastic scheduling of aggregators of plug-in electric vehicles for participation in energy and ancillary service markets. *Energy* 2017;118:1168–79.